

***SURV/SURVMETH 720***  
***Total Survey Error & Data Quality 1***

**Class 2**

**Costs, Coverage of the Target Population**  
**September 8, 2025**

# But first...

Lovely comments on Perusall ! Thank you everyone!

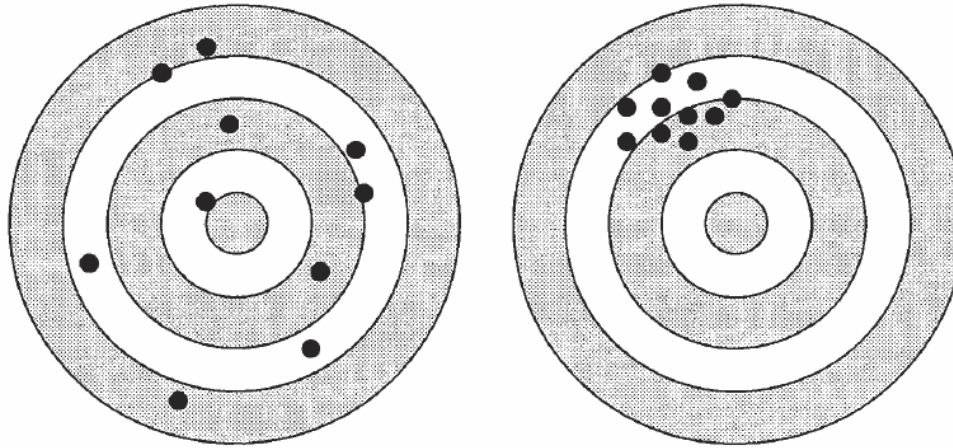
But :

- No “only questions”
- No “only comments”

# Previous class

- Diverse perspectives on survey research
- Overview of TSE framework

# Evaluating 2 designs – concept of an MSE

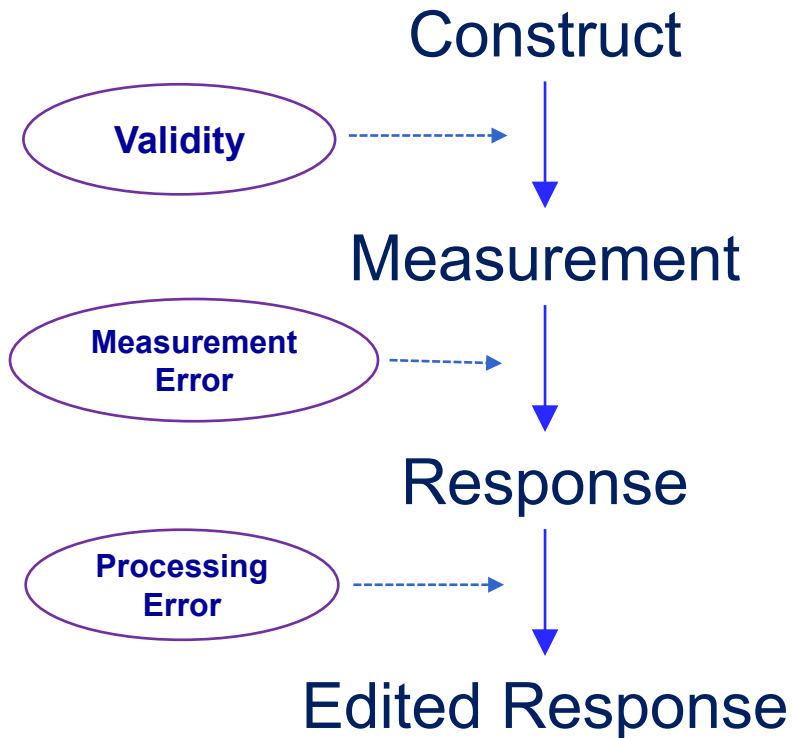


- $MSE = E(\hat{\theta} - \theta)^2$
- $MSE = Bias^2(\hat{\theta}) + Var(\hat{\theta})$
- Bias =
- Variance =

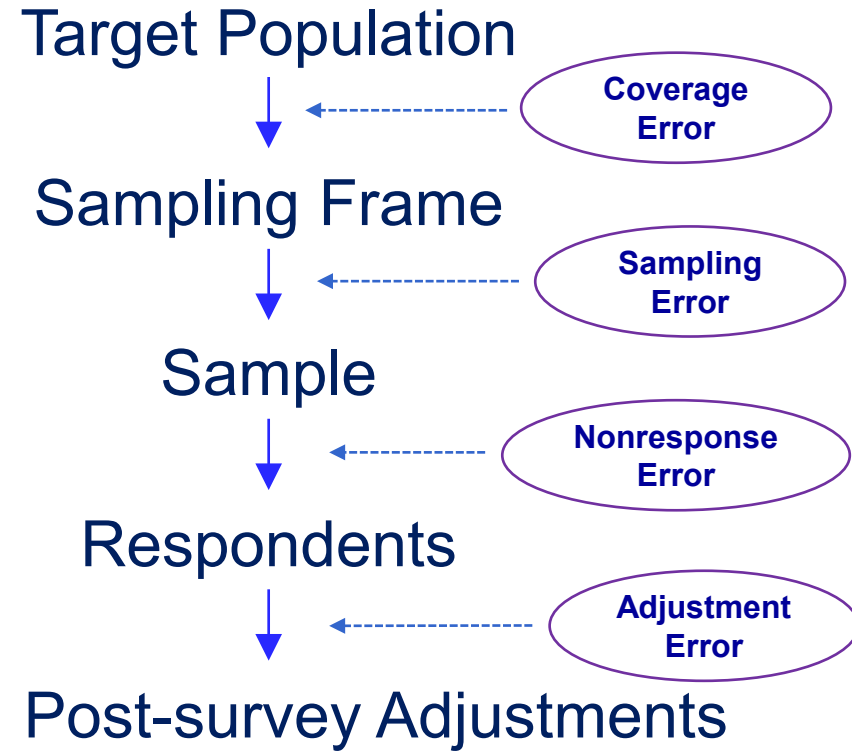
Aria: “Coverage error == Representation error ?”

# TSE Framework

## Measurement



## Representation



Survey Statistic

# Group activity

Conceptually, can all sources of error that are normally associated with bias also be associated with variable error?

i) Yes

ii) No

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## Building Intuition about “Quality”

“I have designed and implemented a survey that produces very accurate estimates.”

Is this enough ?



# “Fitness of Use” of Survey Statistics

| Dimension        | Description                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------|
| Accuracy         | Degree to which error is minimized                                                                         |
| Relevance        | Degree to which it meets the real needs of users                                                           |
| Timeliness       | Delay between the reference point and the date it becomes available                                        |
| Coherence        | Degree to which it is consistent over time and can be successfully brought together with other information |
| Accessibility    | Ease with which it can be obtained                                                                         |
| Interpretability | Availability of metadata and other supplemental information necessary to interpret and utilize it          |

*Brackstone (1999)*

# A System for Product Improvement, Review and Evaluation, ASPIRE

## Statistics Sweden

### Results for the Labour Force Survey, Round 3 in 2013

|                                      | Error Source           | Average score round 2 | Average score round 3 | Knowledge of Risks | Communication | Available Expertise | Compliance with standards & best practices | Plans or Achievement towards mitigation of risks | Risk to data quality |
|--------------------------------------|------------------------|-----------------------|-----------------------|--------------------|---------------|---------------------|--------------------------------------------|--------------------------------------------------|----------------------|
| Accuracy (control for error sources) | Specification error    | 70                    | 70                    | ☺                  | ☺             | ☺                   | ☺                                          | ☺                                                | L                    |
|                                      | Frame error            | 58                    | 58                    | ☺                  | ☺             | ☺                   | ☹                                          | ○                                                | L                    |
|                                      | Non-response error     | 52                    | 52                    | ○                  | ○             | ○                   | ○                                          | ○                                                | H                    |
|                                      | Measurement error      | 56                    | 68                    | ☺                  | ☺             | ☺                   | ○                                          | ☺                                                | H                    |
|                                      | Data processing error  | 62                    | 62                    | ○                  | ○             | ☺                   | ☺                                          | ☺                                                | M                    |
|                                      | Sampling error         | 78                    | 80                    | ☺                  | ○             | ☺                   | ○                                          | ☺                                                | M                    |
|                                      | Model/estimation error | 60                    | 64                    | ○                  | ○             | ☺                   | ☺                                          | ☺                                                | M                    |
|                                      | Revision error         | N/A                   | N/A                   | N/A                | N/A           | N/A                 | N/A                                        | N/A                                              | N/A                  |
| Total score                          |                        | 60,9                  | 64,3                  |                    |               |                     |                                            |                                                  |                      |

| Scores |      |      |           |           | Levels of Risk |        |     | Changes from round 2 |                |
|--------|------|------|-----------|-----------|----------------|--------|-----|----------------------|----------------|
| ☹      | ☹    | ○    | ☺         | ○         | H              | M      | L   |                      |                |
| Poor   | Fair | Good | Very good | Excellent | High           | Medium | Low | Improvements         | Deteriorations |

[https://www.scb.se/contentassets/edf20d31969940249b04d67cebcd5f73/ov9999\\_2020a01\\_br\\_x99br2101.pdf](https://www.scb.se/contentassets/edf20d31969940249b04d67cebcd5f73/ov9999_2020a01_br_x99br2101.pdf)

Related concept: "Quality profile"

# TSE paradigm...

- Part of the broader concept of Total Survey Quality (or even Total Research Quality)
- Focuses on accuracy, that is, reducing MSE

# Weaknesses of the TSE Framework

1. Exclusions of key quality concepts
  - Notably the relevance to the user
2. Diagrammatic representation might give an impression of independence of error sources
3. Lack of routine measurements
  - error/quality profiles are useful but rare
  - empirical estimation of errors is hard to achieve (but can make use of 'effect generalizability'.); large design burden.

# Weaknesses of the TSE Framework...

4. Does not define acceptable error thresholds
5. Ineffective influence on professional standards
  - press releases by many survey organizations rarely mention anything beyond sampling errors (if that)

# Strengths of the TSE Framework

## 1. Decomposition of errors

- nomenclature for different components

## 2. Separation of phenomena affecting statistics in different ways

- measurement vs. representation issues
- variance vs. bias
- provides way to evaluate alternative data collection procedures (trade-offs, resource allocations)

## 3. Conceptual foundation of the field of survey methodology

- subfields defined by errors

# Moving to today's focus

- Costs
- Coverage of the target population

# Costs



# Cost-Error Relationship

- Key contributor to the risk of survey error
- Many efforts to reduce errors increase costs
- This includes the strategies that we will cover throughout this course for:
  - Coverage error reduction
  - Nonresponse error reduction
  - Measurement error reduction
- What about reducing costs – does this always increase error?

# Cost-Error Constraints

- Surveys are conducted with constraints
- Three common types :
  - Set budget. Minimize errors within \$\_\_\_\_.
  - Range of acceptable costs. Explore design options that produce different error levels within cost range.
  - Minimize cost for a given error level, e.g., Sampling variance  $\leq$  \_\_\_\_.

# Traditional Cost Model

- Total cost = Fixed costs + Variable costs.
  - Fixed costs: costs of doing a survey regardless of its design.
  - Variable costs: costs that are a function design features.

# Complicated Characteristics of Survey Costs

- Nonlinearity
  - Cost per case decreases as sample size increases
- Discontinuity
  - Need to add resources for every nth case
- Stochastic features
  - Variability in costs across cases
- Limited domains of applicability
  - Cost structure can change (cost model breaks down) in different organizations or with large design changes
  - Surveys with different lengths

# Steps for Generating Cost-Error Model

1. Specify error model as function of design features.
2. Specify cost model in terms that are shared by the error model.
3. Combine equations into optimization problem that aims to minimize cost/error subject to an error/cost constraint.
  - e.g., determine the sample size and allocation to strata that minimizes error subject to a cost constraint

# Limitations of Cost-Error Models

- Error models are for single estimates, for a single variable, and usually limited to sampling variance. And what about subpopulations?
- Error must be measurable and specified as a function of design features
  - In practice, this is more feasible for sampling error than other error sources
- Difficult to account for complicated characteristics of survey costs
- Major alternative designs have different budgets

Increasing research on costs ...

# Increasing research on costs ...

1. Olson et al. (2021). Survey Costs: Where are We and What is the Way Forward ?

- Typology for survey cost metrics
- Suggest criteria to evaluate the utility of cost metrics:

2. Wagner et al. (2021). Using Time Series Models to Understand Survey Costs.

- Focus on per-interview cost over time in a repeated cross-section survey; treat this as you would a survey estimate (bias/variance)
- Examine trends and seasonality; remainder is process variability
- Examine causes of error, e.g., incorrectly assigning hours worked on a specific task to the wrong task



# Increasing research on costs ...

## 3. Wagner et al. (2024). Proxy Survey Cost Indicators in Interviewer-Administered Surveys: Are they Actually Correlated with Costs?

- How do proxy indicators of cost correlate with actual cost ?
- Total attempts, total contacts, total completes, and sample size are strongly correlated with total interviewer hours (average correlation 0.82) and moderately with total costs (average correlation 0.60).
- Indicators may provide important signals

4. Olson et al (2024). Examining Variation in Survey Costs Across Surveys.

5. Coffey, S. M., and M. R. Elliott. 2023. “Optimizing Data Collection Interventions to Balance Cost and Quality in a Sequential Multimode Survey.”

6. Wagner et al (2025). Using a Stopping Rule to Optimize Cost-Quality Tradeoffs in a Large, Mixed-Mode Survey: A Simulation Study.

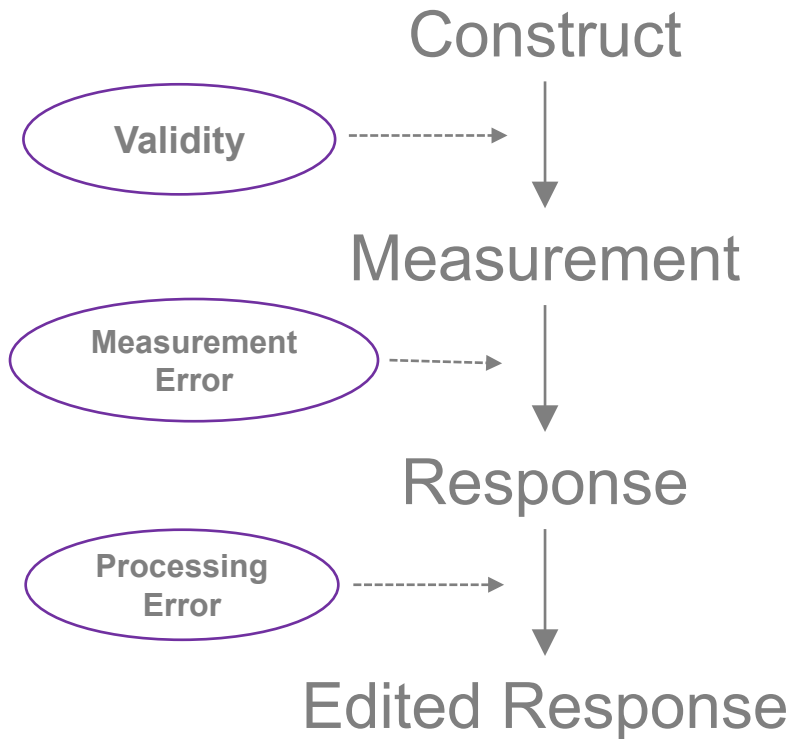
“The challenge is to produce the highest quality data possible within a fixed budget. We use a stopping rule designed to minimize a function of cost and errors. The rule is based on the product of predicted costs and the predicted mean squared error of a survey estimate.”

# Moving to Coverage Error

(Remember the three things you need to ask about every error source ?)

# TSE Framework

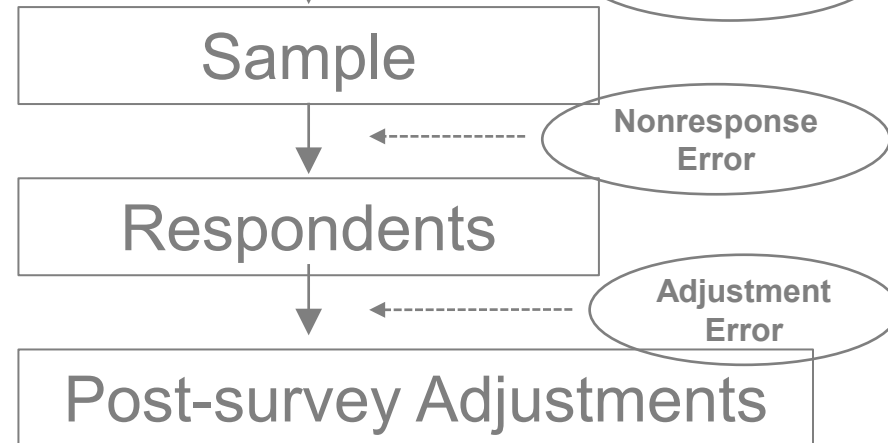
## Measurement



## Representation

**Target Population**

**Sampling Frame**



Survey Statistic

# Coverage Error

- Arises because some members of the **target population** are not represented on the **sample frame**

# From Kish 1965 (p.7)

Finite population defined in terms of:

1. Content

“all civilians”

2. Units

“in family units”

3. Extent

“in the lower 48 states and District of Columbia in the U.S.”

4. Time

“in 2024”

The definition of the term *population* is not standardized in the field. Some authors use different terminology to define each of these groups. For example, many textbooks use the term *target population* for *population of inference* used here. *Sampled population* is used to refer to the target population. Careful attention should be given to the definitions used by the authors of survey documents.

*James M. Lepkowski*

- Lots of comments on this (Grace, Aria, Brittany, Zhenjing, Yuchen):

“Because of the traditional concerns in survey research with inference to groups external to the sample cases themselves (in contrast to the concerns with internal control of experimental research), there have been some distinctions among different types of population (some of them only abstract concepts) relevant to the survey.”



# Incidentally, another reason not to trust AI blindly...

**Abstract Population:** "All young adults in the United States."

- This is an abstract population because there is no perfect, comprehensive list of every single young adult in the U.S. at any given moment. People are constantly aging into and out of the category, moving, or changing their contact information.
- You can't get a list of names and addresses for all of them, so you have to work with a sample.

**In contrast, a "concrete population" would be:**

- **Example 1:** "All students currently enrolled at a specific university." You could obtain a complete list of these students from the registrar's office.
- **Example 2:** "All members of the American Academy of Health Behavior." The organization would have a directory or database of all their members.

You'll see different terms used (clarify!):

- Inferential population (Want results from a sample survey to be generalized to this. Conceptual.)
- Target population (Set of elements to be studied. Time restrictions. Observable.)
- Frame population (Set of TP members that have a chance to be selected into the sample. 'Operational'.)
- 'Survey population' ("Set of FP who might be Rs". Accessible. Practical.)

# “Hidden coverage issues” (Doris, Nicholas, Zhenjing, Thomas, Lingchen, Tavril, Jeremy)

- but note that coverage error is really the discrepancy between the Target population and the Frame population (and not the survey population)

-

- Is “inferential population” == “superpopulation” ?  
(Joy’s comment)

Furthermore, sample data are also used for inferences beyond the target populations, and these are many, various, and ill defined. “Superpopulations” of sampling theory are not only among these, but behind all these inferential populations. These model-dependent inferences (1.8) are too often merely implicit. Even more vagueness describes the path of judgment samples

Kish, Statistical design for research, p.29

- What about the term “universe” ?

# Sample Frame

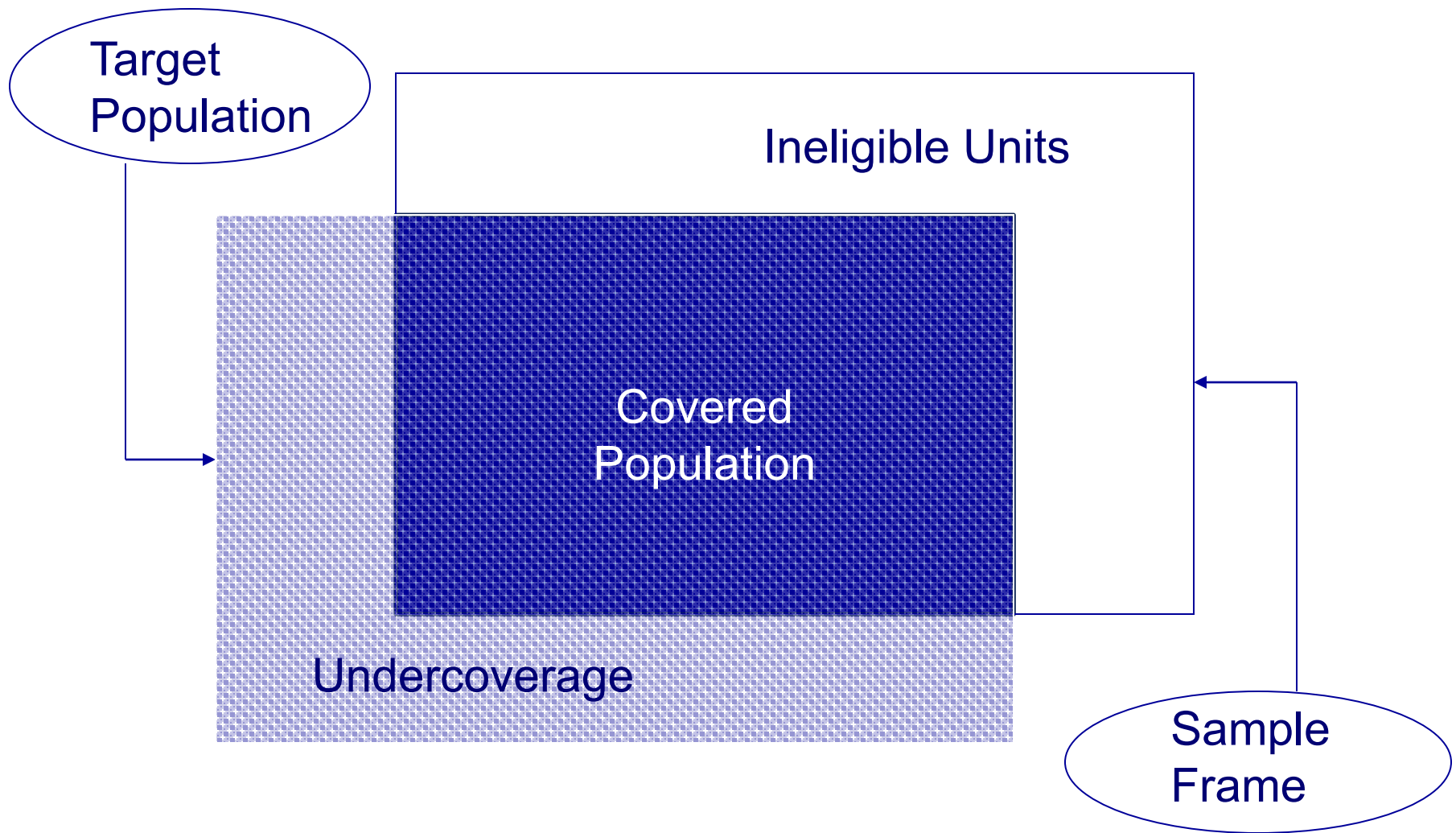
- Lists/methods/procedures used to identify units in the target population
- Components:
  - Identifiers that distinguish the units (names, ID numbers)
  - Information that can provide access to units (such as addresses, telephone numbers, e-mail addresses)

# Mappings between Frame and Target Population

| Frame | Target Population |                              |
|-------|-------------------|------------------------------|
| F     | T                 | Correspondence               |
| .     | T                 | <b>Undercoverage</b>         |
| F     | .                 | Foreign elements/Ineligibles |
| F, F  | T                 | Multiplicity (many-to-one)   |
| F     | T, T              | Clustering (one-to-many)     |

“If Virginia and DC cellphone sampling frames are also used to sample for Maryland, the undercoverage percentage for all Maryland users will drop from 21.5 to 14.1%. However, the overcoverage will increase from 19.3 to 66.3%.”

*Marken et al. 2019 (“Geographic Inaccuracy of Cellphone Samples”)*





# Types of Frames in Common Use

- Area frame
- Address-based frames
- RDD frames
- Register frames

# Area Frames

- Collection of geographic units
- Area sampling process:
  - Define geographic unit (e.g., counties, metropolitan areas, zip code areas, sections from satellite image)
  - Randomly select areas
  - List units within those areas
  - Select subset of housing units
- Used primarily for in-person surveys

# Address-Based Frames

- List of household addresses
  - Based on Postal Service's Computerized Delivery Sequence (CDS) file, which provides nearly 100% coverage in many areas and is updated regularly
- ABS process:
  - Acquire address list
  - If in-person survey: select geographic areas
  - Select housing units from selected geographic areas
- Used for in-person interviews and surveys that start with a mail invitation, including “web-push” surveys

# Characteristics of Area Frames and ABS

## Area Frame

- On-site enumeration by listers (?)
- Geographic location is known
- Potential coverage issues:
  - subject to listing errors
  - hard-to-see housing units

## ABS

- No listing required (less expensive)
- Geographic information not known (can be added through geocoding)
- Potential coverage issues:
  - newly constructed housing units
  - households' P.O. boxes
  - some units are businesses, not residences

# RDD Frame

- Randomly-generated telephone numbers
- **Random Digit Dialing (RDD)** process:
  - Define “banks” of 100 phone numbers with the same area code and next 5 digits (XXX-XXX-XX00 through XXX-XXX-XX99)
  - Identify banks with some minimum number of listed residential numbers (at least for landline RDD)
  - Randomly call phone numbers in these banks

# Register Frame

- Administrative list of units compiled for some other purpose
  - UMD/UMICH alumni database; AAPOR member database; Census Business Register; IRS list of registered nonprofits; NASS list frame of agricultural producers; NCES list frame of private schools
  - Quality depends on the list and how it is maintained
    - Frequency of new units being added
    - Is contact information updated regularly?
    - Are duplicates removed?

# Measuring Coverage Error

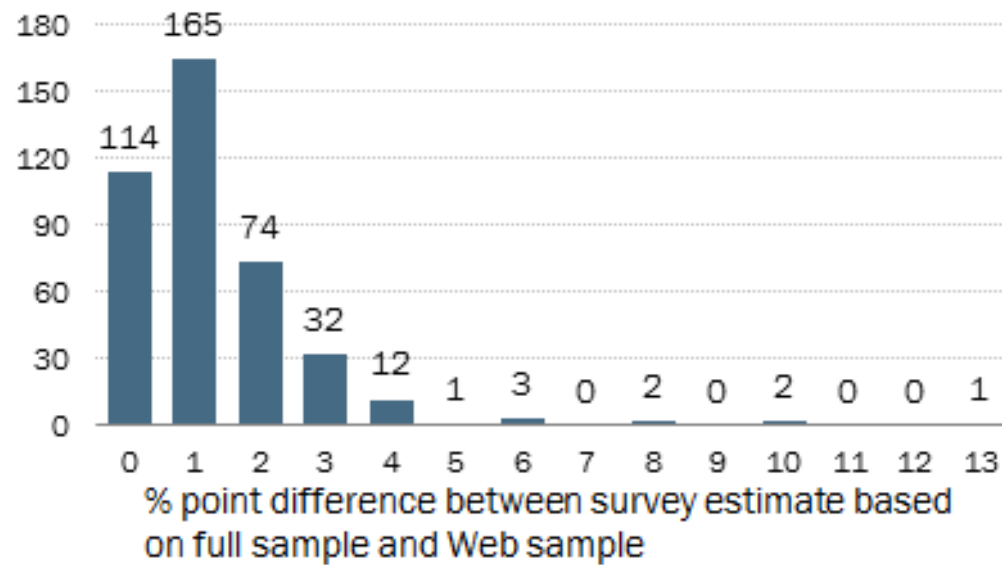
What is this “error”?

# Coverage Bias

For a mean, error due to undercoverage can be decomposed into terms for the proportion of target population units that are not covered (U) **AND** on how different these units are from the units that are covered (C)

$$\bar{Y}_C - \bar{Y} = \frac{U}{N} (\bar{Y}_C - \bar{Y}_U)$$





Source: 406 survey estimates from the American Trends Panel Waves 3-11 conducted during 2014 and 2015

PEW RESEARCH CENTER

<https://www.pewresearch.org/methods/2015/09/22/coverage-error-in-internet-surveys/>

$$U/N \approx 0.18$$

# But...

*Demographic profiles of all respondents, Web respondents and mail respondents*

|              | <b>Total Sample</b><br><i>Unweighted</i> | <b>Web Sample</b><br><i>Unweighted</i> | <b>Mail Sample</b><br><i>Unweighted</i> |
|--------------|------------------------------------------|----------------------------------------|-----------------------------------------|
|              | %                                        | %                                      | %                                       |
| Has internet | 94                                       | 100*                                   | 48                                      |
| Male         | 48                                       | 49                                     | 44                                      |
| Female       | 52                                       | 51                                     | 56                                      |
| 18-29        | 13                                       | 14*                                    | 1                                       |
| 30-49        | 27                                       | 29*                                    | 9                                       |
| 50-64        | 33                                       | 33                                     | 34                                      |
| 65+          | 27                                       | 24                                     | 56*                                     |

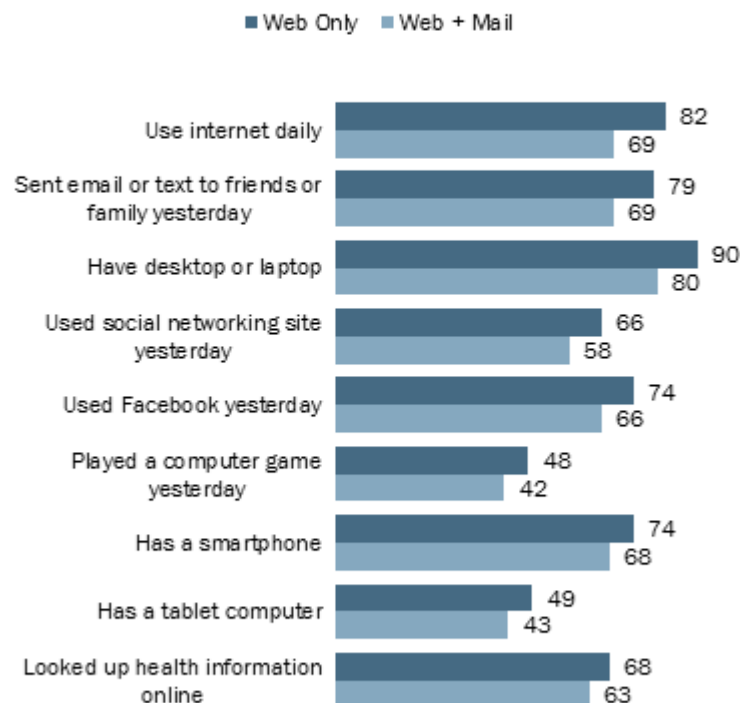
\* Statistically significant difference between Web and mail respondents.

Note: Mail respondents include adults who do not have internet access or have internet access but do not have or would not provide an email address to participate via the Web.

Source: Pew Research Center Panel survey wave 11, June 2-29, 2015

## Technology and Internet Items Show Sizeable Differences

*Comparisons of Web-only and total sample, in %*



Source: American Trends Panel Waves 5, 8 and 9.

PEW RESEARCH CENTER

# Group Activity - Instructions

- Go to: [https://bit.ly/720\\_2025\\_Class2](https://bit.ly/720_2025_Class2)
- Find the slide corresponding to your group
- Introduce yourselves and write your names on the slide
- Designate a volunteer who could speak about the slide when your group is called

# Measuring Undercoverage

- Most of the time the researcher has limited means of measuring coverage error
  - Both the noncoverage rate and difference between the noncovered and covered populations are unknown
- One strategy is to compare data from frame to external data (assumed to be better), e.g., Census data or data from another frame

# Measuring Undercoverage...

- Another strategy is analyze data from large-scale survey where membership on a frame (that was not used by the survey) is measured
  - Internet access measured in area frame survey
  - Phone access measured in ABS survey

# Comparing Bias across Variables

- When estimating bias for different variables, the units will likely vary and this makes it challenging to make meaningful comparisons
- Can rescale into relative bias ( $\frac{B}{\bar{y}}$ )

# Next Week's Class

- Repairs for undercoverage
- Reading assignment #2
- Questions?